

9.2 Wave Equation

When a transverse wave is passing over the surface of water, the particles of water move up and down about their mean position. As a matter of fact each particle executes simple harmonic motion, only the different particles pass through their positions of maximum displacement one after the other. Such a wave is known as a **simple harmonic wave**. Suppose a particle O moving in a medium with a constant velocity v . Each particle of the

medium is affected a little later than the particle preceding it and this delay, *i.e.*, the phase difference between successive points from point to point along the wave. The solid line curve in Fig. 9.12 represents the displacements

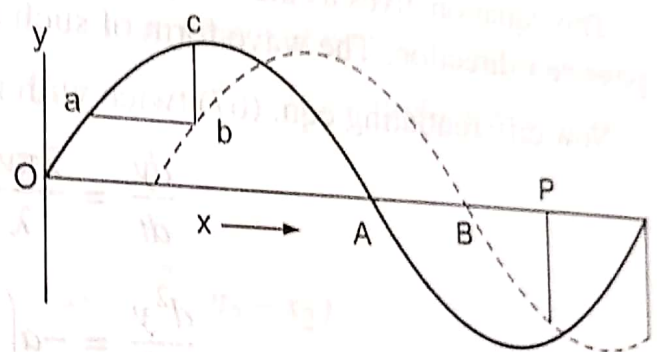


Fig. 9.12 Illustrating relationship between wave velocity and particle velocity

of all the particles at any instant regarded as zero and the dotted line curve the position of the wave at time t later. If we measure time from the movement that the particle O passes through its mean position in the +ve direction, its displacement after time t along the line Oy is given by

$$\begin{aligned} y &= a \sin \omega t \\ &= a \sin \frac{2\pi t}{T} \end{aligned} \quad \dots(63)$$

where a is the amplitude of the vibrating particle and T is the periodic time, that is the time taken by the wave to travel a distance λ .

The displacement at the same instant of a particle P at a distance x from O towards the right can be written as

$$y = a \sin (\omega t - \phi) \quad \dots(64)$$

where ϕ is the phase lag as particle P is situated towards the right of O .

Let us evaluate the phase constant ϕ in terms of the wavelength and the distance x of the particle P from O . We know that a change of 2π in phase occurs between two particles separated by a distance λ , hence a change ϕ in phase corresponding to a distance x , is given by

$$\phi = \frac{2\pi x}{\lambda} \quad \dots(65)$$

Therefore, we obtain

$$\begin{aligned} y &= a \sin \left(\omega t - \frac{2\pi x}{\lambda} \right) = a \sin \left(\frac{2\pi t}{T} - \frac{2\pi x}{\lambda} \right) \\ \text{or} \quad y &= a \sin 2\pi \left(\frac{t}{T} - \frac{x}{\lambda} \right) \end{aligned} \quad \dots(66)$$

This is one form in which the equation of wave motion of the simplest type is represented. Also $v = \frac{\lambda}{T}$, where symbols have their usual meanings. Thus, we can write above equation as

$$\begin{aligned} y &= a \sin 2\pi \left(\frac{vt}{\lambda} - \frac{x}{\lambda} \right) \quad \left(\because v = \frac{\lambda}{T} \right) \\ \text{or} \quad y &= a \sin \frac{2\pi}{\lambda} (vt - x) \end{aligned} \quad \dots(67)$$

This equation gives us the complete form of a wave of amplitude a propagating with velocity v , in the positive x direction. The wave form of such an equation is shown in Fig. 9.13

Now differentiating eqn. (67) twice with respect to time t , we have

$$\begin{aligned} \frac{dy}{dt} &= \frac{2\pi va}{\lambda} \cos \frac{2\pi}{\lambda} (vt - x) \\ \text{and} \quad \frac{d^2y}{dt^2} &= -a \left(\frac{2\pi}{\lambda} \right)^2 v^2 \sin \frac{2\pi}{\lambda} (vt - x) \end{aligned} \quad \dots(68)$$

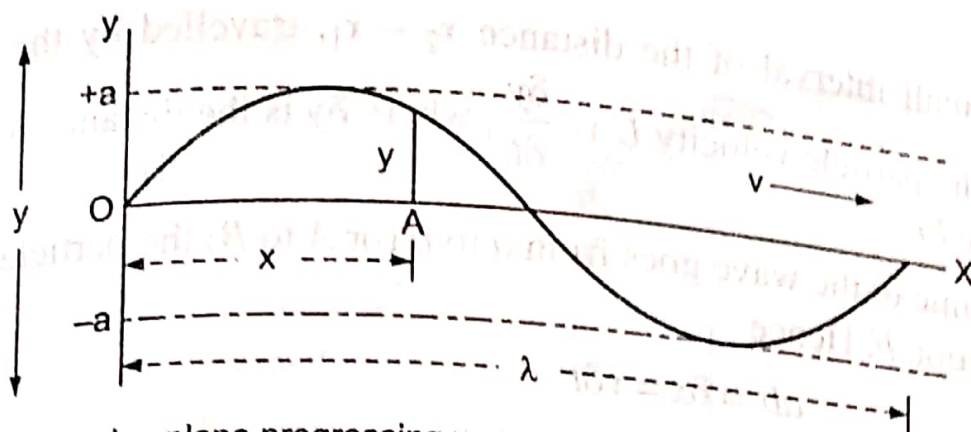


Fig. 9.13 Showing plane progressing wave moving with velocity v in the $+x$ direction

Also differentiating eqn. (67) twice with respect to x , we get

$$\frac{dy}{dx} = -\frac{2\pi a}{\lambda} \cos \frac{2\pi}{\lambda}(vt - x)$$

$$\frac{d^2y}{dx^2} = -a \left(\frac{2\pi}{\lambda} \right)^2 \sin \frac{2\pi}{\lambda}(vt - x)$$

Comparing above equation with eqn. (68), we get

$$\frac{d^2y}{dt^2} = v^2 \frac{d^2y}{dx^2}$$

This is the differential equation of the wave motion.

From eqn. (67) we can find the phase difference between two particles P_1 and P_2 at distances x_1 and x_2 from particle O respectively. Therefore, the displacement for P_1 is

$$y_1 = a \sin \frac{2\pi}{\lambda}(vt - x_1)$$

$$y_2 = a \sin \frac{2\pi}{\lambda}(vt - x_2)$$

The phase difference ϕ is

$$\phi = \frac{2\pi}{\lambda}(vt - x_1) - \frac{2\pi}{\lambda}(vt - x_2) = \frac{2\pi}{\lambda}(x_2 - x_1) = \frac{2\pi}{\lambda} \times \text{path difference}$$